MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-29A/U

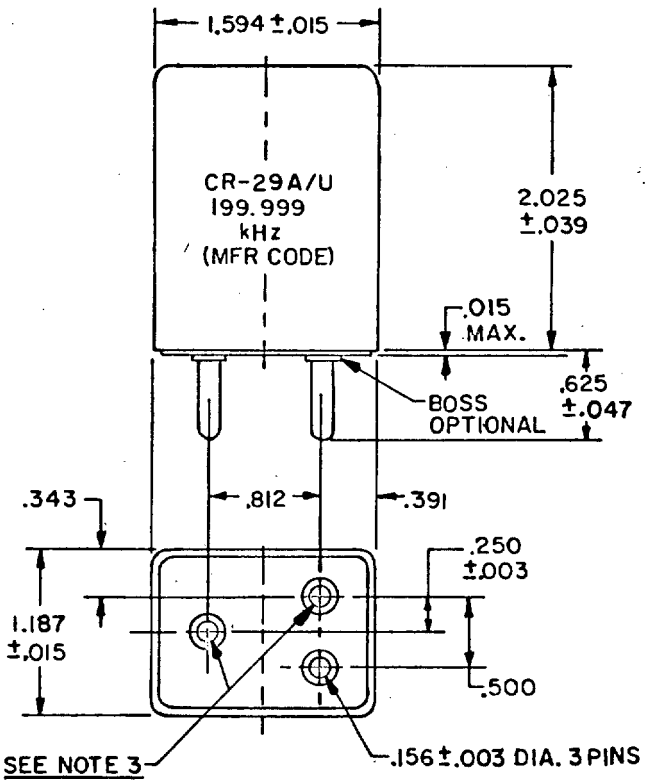
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of specification MIL-C-3098.

Pertinent characteristics: 80 to 200 kHz; fundamental; controlled; antiresonance.

Ⓓ THIS SPECIFICATION INACTIVE FOR NEW DESIGN

INCHES	MM	INCHES	MM
.003	.08	.391	9.93
.005	.13	.500	12.70
.015	.38	.625	15.88
.039	.99	.812	20.62
.047	1.19	1.187	30.15
.156	3.96	1.594	40.49
.250	6.35	2.025	51.44
.343	8.71		



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type-designation, frequency, and code-designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
4. Unless otherwise specified, tolerance is ±.005 (.13 mm).

FIGURE 1. Dimensions and configuration.

Ⓓ denotes changes.

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-21/U (see MIL-H-10056/10)
Dimensions, marking, and configuration	See figure.
Frequency range	80 to 200 kHz, inclusive
(D) Frequency tolerance, operating temp. range	±20 PPM
(D) Frequency stability	±5 PPM
Equivalent resistance	See table II
Antiresonance, load capacitance	32.0 ±0.5 pF
Mode of oscillation	Fundamental
Reference temperature	75° ±1°C
Temperature ranges:	
Operating	70° to 80°C, inclusive
Operable	-40° to +70°C, inclusive
Test set, calibration values and rated drive level	See table III
Shock:	
(D) Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Vibration 1/:	
(D) Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
(D) Thermal shock:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Bond strength: kHz	Grams, minimum
80 to 120	1,000
120+ to 160	800
160+ to 200	700

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (0.030 inch total excursion).

TABLE II. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
kHz	Ohms
80 to 120	5,000
120+ to 160	5,500
160+ to 200	6,000

TABLE III. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	ohms	mA	volt	mW
TS-710/TSM	80 to 120	3,500	---	2.65	2.0 ± 0.4
	120+ to 160	3,900	---	2.8	2.0 ± 0.4
	160+ to 200	4,200	---	2.9	2.0 ± 0.4

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NOTES:
1. This crystal unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.
2. Qualification is not required. First article inspection is applicable.
3. Pin alinement and frequency and resistance at room temperature (for controlled types) requirements are not applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army -
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5955-0512-4)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3098/11D CRYSTAL UNITS, QUARTZ, CR-29A/U

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

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DATE

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EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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